

University of Colombo School of Computing (UCSC)

University of Colombo

SCS 1307 – Probability and Statistics

Assignment 02

Semester I – 2026

Index Number: _____

Instructions:

1. Answer all questions.
2. Show all necessary steps in your calculations
3. Non-programmable calculators are allowed
4. Duration: 1 hour

Q1. If $X \sim \text{Binomial}(10, p)$ and the mean of the distribution is 6,

(a) Find the probability of success. (4 marks)

(b) Hence, calculate the probability that exactly 7 successes occur. (8 marks)

Q2. A database server experiences an average of 2 failures per day. Assume that failures occur independently.

Calculate the probability that at least one failure occurs on each of the next two days (12 marks)

Q3. The download speed experienced by users of an internet service provider is approximately normally distributed with a mean of 120 Mbps and a standard deviation of 15 Mbps.

Determine the value of k such that 90% of users experience download speeds between $120 - k$ Mbps and $120 + k$ Mbps. (12 marks)

Q4. A cloud server receives an average of 100 API requests per minute.

Using the normal approximation with continuity correction, calculate the probability that between 90 and 110 requests (inclusive) arrive during the next minute. (12 marks)

Q5. A machine learning classifier correctly identifies an image with probability 0.95. A random sample of 200 images is tested independently.

Using the normal approximation with continuity correction, calculate the probability that between 180 and 195 images (inclusive) are classified correctly. (16 marks)

Q6. The response time of a cloud-based application is approximately normally distributed with a mean of 250 milliseconds and a standard deviation of 40 milliseconds.

Calculate:

(a) the probability that the response time is less than 300 milliseconds. (6 marks)

(b) the probability that the response time lies between 220 and 280 milliseconds. (10 marks)

Q7. A web server receives requests from users at an average rate of 6 requests per minute. Assume that the number of requests arriving during any one-minute interval follows a Poisson distribution.

Calculate the probability that fewer than 4 requests arrive during the next minute. (20 marks)

..... End